PROBLEM SET 1, PART 2: SMOOTH MANIFOLDS/MAPS/FUNCTIONS
DUE: SEP. 26

(1) [**Construct smooth manifolds by gluing Euclidean open sets**]
Let M be a smooth manifold with atlas $\mathcal{A} = \{(\varphi_\alpha, U_\alpha, V_\alpha)\}$.
(a) Prove: the transition maps $\varphi_{\alpha\beta} := \varphi_\beta \circ \varphi_\alpha^{-1}$ satisfy the *cocycle conditions*
(i) $\varphi_{\beta\gamma} \circ \varphi_{\alpha\beta} = \varphi_{\alpha\gamma}$ on $\varphi_\alpha(U_\alpha \cap U_\beta \cap U_\gamma)$.
(ii) $\varphi_{\alpha\alpha} = \mathrm{Id}_{V_\alpha}$.
(iii) $\varphi_{\alpha\beta} = (\varphi_{\beta\alpha})^{-1}$.
(b) Now on the disjoint union $\widetilde{M} := \bigsqcup_\alpha V_\alpha$ we define an equivalence relation via

$$x \sim y \iff \exists \alpha, \beta \text{ s.t. } x \in V_\alpha, y \in V_\beta \text{ and } y = \varphi_{\alpha\beta}(x).$$

(i) Check: $\sim$ is an equivalence relation on $\widetilde{M}$.
(ii) Prove: the quotient $\widetilde{M}/\sim$ is homeomorphic to M.
(iii) Define a natural smooth structure on $\widetilde{M}/\sim$

(2) [**Product of manifolds**]
(a) Prove: The product of two topological manifolds is a topological manifold.
(b) Let M, N be smooth manifolds with atlases $\{(\phi_\alpha, U_\alpha, V_\alpha)\}$ and $\{(\psi_\beta, X_\beta, Y_\beta)\}$ respectively. Define an atlas on the product space $M \times N$ to be $\{(\phi_\alpha \times \psi_\beta, U_\alpha \times X_\beta, V_\alpha \times Y_\beta)\}$. Check that this gives a smooth structure on $M \times N$.
(c) Explicitly construct local charts on $\mathbb{T}^2 = S^1 \times S^1 \subset \mathbb{R}^4$ to make it into a smooth manifold. (You need to prove that your charts are compatible.)

(3) [**The need of global continuity in the definition of smoothness**]
(a) Define two atlases $\mathcal{A}$ and $\mathcal{B}$ on $\mathbb{R}$ as follows:

$$\mathcal{A} = \{(\varphi, \mathbb{R}, \mathbb{R})\}, \quad \varphi(x) = x,$$
$$\mathcal{B} = \{(\psi_i, X_i, Y_i) : i \in \mathbb{Z}\}, \quad \text{where } X_i = Y_i = (i-1, i+1) \text{ and } \psi_i(x) = x.$$

Define a map $f : (\mathbb{R}, \mathcal{A}) \to (\mathbb{R}, \mathcal{B})$ via

$$f(x) = \begin{cases} 2, & x > 0 \\ 0, & x \le 0 \end{cases}$$

Prove: For each i, the map $\psi_i \circ f \circ \varphi^{-1} : \varphi(\mathbb{R} \cap f^{-1}(X_i)) \to \psi_i(X_i)$ is smooth. [So we can't drop the continuity assumption in Definition 1.1 in Lecture 4.]
(b) Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds. Suppose for each point p, there exists $(\varphi_\alpha, U_\alpha, V_\alpha) \in \mathcal{A}$ and $(\psi_\beta, X_\beta, Y_\beta) \in \mathcal{B}$ so that $p \in U_\alpha$ and $f(U_\alpha) \subset X_\beta$. Prove: Under this assumption we can drop the continuity assumption of f in Definition 1.1 in Lecture 4.

(4) [**Orientability of smooth manifolds**]
Let M be a smooth manifold of dimension n. We say M is *orientable* if there exists an atlas $\{(\varphi_\alpha, U_\alpha, V_\alpha)\}$ of M so that

$$\det(d\varphi_{\alpha\beta})_p > 0$$

for all $p \in \varphi_\alpha(U_\alpha \cap U_\beta)$, where $d\varphi_{\alpha\beta}$ is the Jacobian matrix of $\varphi_{\alpha\beta}$.

(a) Prove: S^n is orientable.
(b) Prove: For $\mathbb{RP}^n$, the atlas we constructed in Lecture 2 makes $\mathbb{RP}^n$ orientable for odd n but not for even n.
(c) [Not required] Prove: $\mathbb{CP}^n$ is orientable.
(d) Prove: If M, N are orientable, so is $M \times N$. Conversely, if $M \times N$ are orientable, so are M and N.

(5) [**Smooth Urysohn's Lemma**]
Suppose A and B are two disjoint closed subsets of a smooth manifold M.

(a) Show that there exists a smooth function f on M so that $0 \leq f(x) \leq 1$ for all $x \in M$, $f|_A = 0$ and $f|_B = 1$.
(b) [Not required] For any closed subset K in M, there is a smooth nonnegative function $f : M \to \mathbb{R}$ so that $f^{-1}(0) = K$.
[Hint: First work out this problem for $M = \mathbb{R}^n$, in which case $\mathbb{R}^n \setminus K$ can be written as the union of countably many open balls. Try to construct a smooth function on each such ball which is positive inside the ball and equals 0 outside the ball. Then add these functions (each multiplied by suitable factors to guarantee smoothness).]
(c) Show that there exists a smooth function f on M so that $0 \leq f(x) \leq 1$ for all $x \in M$, $f^{-1}(0) = A$ and $f^{-1}(1) = B$.

(6) [**Exhaustion function**]
A real-valued continuous function f on M is called an *exhaustion function* for M if for any $c \in \mathbb{R}$, the *sublevel set* $f^{-1}((-\infty, c])$ is compact. Prove: There exists a positive smooth exhaustion function on any smooth manifold.